

Mehul Agarwal

Backend Eng | Data Eng | AIMA/GenAI Eng

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SUMMARY

CS graduate (2026) who ships AI-powered systems end-to-end – LLM/RAG multi-agent platforms, vector-search matching engines, and production data pipelines across 7+ source systems. Fluent in LangGraph, RAG, MCP, pgvector, and Azure OpenAI; 700+ DSA problems solved.

WORK EXPERIENCE

Navikenz

Mar 2026 – Present

Software Engineer Intern, Data Engineering and Backend

Noida, India

Engineered a multi-source **data ingestion framework** moving data from **7 external source types** (Snowflake, SQL Server, Azure SQL, Oracle, ADLS Gen2, SharePoint, Amazon S3) into a Fabric Lakehouse through a bronze → silver → gold pipeline, cutting manual data-transfer effort by **60%**.

Built a FastAPI control-plane that triggers Fabric notebooks for ingestion and rollback, with Key Vault-backed secret resolution (credentials never touch the app) and CDC-style upserts via primary-key columns.

Instrumented every run with row/null counts, completeness %, and batch-level audit & rollback (`batch_id`), reducing ingestion errors by **30%** and resolving **100%** of reported data-flow incidents within SLA.

Pristine Business Solutions

May 2025 – Jul 2025

Software Developer Intern, Enterprise Systems and Data Validation

Noida, India

Implemented automated data validation pipelines that improved workflow reliability by **35%** across enterprise production systems.

Optimized complex SQL queries through index tuning and query restructuring, reducing average execution time by **40%** on large datasets.

PROJECTS

FormatIQ – AI Multi-Agent Document Reformatting Platform

GitHub Live

Scoped, built, and deployed an AI multi-agent platform with **3 domain-agnostic workflows** (document generation, style transfer, compliance checking) using LangGraph and Azure OpenAI; maps draft content into a template via RAG and surfaces a side-by-side diff for human review. Built a FastAPI backend orchestrating multi-agent pipelines with Neon Postgres, Qdrant vector search, and Server-Sent Events for real-time streaming; React (Vite + Tailwind) frontend exports final reports to `.docx/.pdf`.

Tech Stack: Python, FastAPI, LangGraph, RAG, Azure OpenAI, React, Docker, Qdrant, PostgreSQL

CareerMatch AI – LLM-Powered Resume-to-Job Matching Engine

GitHub

Built an end-to-end pipeline that extracts and parses resumes (PDF/DOCX, OCR fallback) into strict structured JSON via LLM, embeds them (384-dim, sentence-transformers) and matches them against jobs scraped from **3 live sources** (Adzuna, Remotive, Arbeitnow) using pgvector HNSW similarity search.

Designed a funnel-shaped matching engine: vector search top-40 → hard filters (location/salary/experience) → top-15 → LLM-based skill-gap/ATS keyword scoring → blended score ($0.6 \times \text{similarity} + 0.4 \times \text{skill overlap}$).

Architected a FastAPI backend on Neon Postgres (relational data + pgvector, single database) with a provider-agnostic LLM router, JWT auth, rate limiting, and Alembic migrations; roadmap includes AI resume tailoring and auto-apply.

Tech Stack: Python, FastAPI, OpenAI, sentence-transformers, pgvector, Neon Postgres, JWT, Alembic

Sovaria – Authority-Driven Backend-as-a-Service Platform

GitHub Live

Constructed a distributed Backend-as-a-Service spanning **11 specialized engines** (identity, policy, NoSQL/SQL storage, temporal versioning, realtime, simulation, functions, billing) behind a single Identity → Policy → Execution pipeline.

Delivered low-latency WebSocket realtime (5–20 ms from commit) and shipped identical Python/TypeScript SDKs plus a CLI with dry-run simulation for CI/CD.

Tech Stack: Python, TypeScript, Distributed Systems, NoSQL/SQL Engines, WebSockets, REST API

Code Similarity and Plagiarism Detection System

GitHub Live

Built a plagiarism-detection tool using AST parsing, tokenization, and control-flow analysis to flag structural code similarity across large submission sets, with an interactive dashboard ranking suspicious pairs for review.

Tech Stack: Python, Flask, Algorithms, SQL

SKILLS

Languages: Python, TypeScript, C++, SQL, JavaScript

AI/ML and GenAI: LLMs, LangChain, LangGraph, RAG, MCP, Agent Frameworks, Azure OpenAI, Vector Databases, Prompt Engineering, TensorFlow

Web and Backend: FastAPI, Node.js, Express.js, React.js, REST APIs, WebSockets, Server-Sent Events

Data Engineering: ETL/ELT Pipelines, Apache Spark, Microsoft Fabric, Snowflake, Data Modeling, Data Quality, Pandas, NumPy

Cloud and DevOps: Microsoft Azure, AWS (S3), Docker, CI/CD, Git, GitHub

Databases: PostgreSQL, MySQL, MongoDB, Qdrant, pgvector

EDUCATION

Jaypee Institute of Information Technology

2022 – 2026

Bachelor of Technology, Computer Science and Engineering

Noida, India

CGPA: 8.2 / 10

LEADERSHIP AND COMMUNITY

Mentored junior students in Data Structures and Algorithms, breaking down problem-solving patterns and guiding hands-on practice.

Lead Coordinator, Optica Club (2025): scoped and ran a campus coding contest end-to-end – problem selection, logistics, and prize-money distribution.